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Krishnakumar Muthukrishnan
Portfolio Website

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Objective

- Objective of this resume is to provide a small insight into the technical capabilities I gained through my experience as a telecommunications professional with more than 6 years' experience working for **Research & Development** wing of a French global Telecommunication organization **Alcatel-Lucent** for building and maintaining their legacy PABX system and the exposure I gained through my latest endeavor of pursuing a Master's degree.(**click here for detailed profile**)

Education

MASTER OF SCIENCE – COMPUTER SCIENCE & ENGINEERING

STATE UNIVERSITY OF NEW YORK – BUFFALO, NEW YORK

- Expected graduation 12/2022
- CGPA - 3.29
- **Relevant Courses** - Machine Learning, Algorithm Analysis, Parallel & Distributed Computing, Software Security, Modern Network Concepts, Android Malware Detection (Java) (**click here for complete list**)

BACHELOR OF TECHNOLOGY: INFORMATION TECHNOLOGY

THIAGARAJAR COLLEGE OF ENGINEERING,ANNA UNIVERSITY – MADURAI, INDIA

- Graduation - 04-2015
- CGPA - 3.33

Experience

2021-08 - Current
(non-technical part time)

Student Employee, Campus TEES, Campus Dining & Shops

University at Buffalo, North Campus

2019-07 – 2021-06

Information Technology Analyst

Tata Consultancy Services, Chennai, India

2015-06 - 2019-06

System Engineer / Software Development Engineer

Tata Consultancy Services, Chennai, India

Technical Skills & Abilities (click here for detailed view)

Languages : C,C++,Python, Javascript,GNU Bash, Solidity, ReactJS/React-Native, HTML,CSS, SQL, Firebase,Java

Framework/Libraries/Protocols : TCP/IP,UDP,HTTPS,SNMP,DHCP,RTP ,SSL/TLS, DTLS, OpenMP, OpenMPI, Pandas, Numpy, POCO

Platform & virtualization environment : Windows, Unix/Linux, VmWare,KVM

Tools : VSCode, RemixIDE,GitHub, Mercurial, Jenkins, Putty, WinSCP, AWS, Wireshark, Android studio, JIRA, VmWare Esxi,KVM Manager,

Leadership : Lead multiple small scale development activities over the span of 6 years.

Notable Projects: (*** click here for elaborate view)

(TCS-ALE) DTLS based End-to-End Security Architecture:(*** click here for elaborate view)

- 100% software based security native implementation replacing the HSMs(Hardware Security Modules)
- Development involved incorporation of the encryption modules in the existing design using the Object Oriented Programming approach.

(TCS-ALE) Cloud migration of legacy PABX system :(*** click here for elaborate view)

- I was part of a development team that worked on a In short I was working for the client side implementation which maintains, validates and does a periodic license checks of the products deployed by the client through communication with the cloud servers deployed.
- The techstack includes C++ with extensive usage of the XMPP protocol for instant secure communication to work over the Linux platform.

(TCS-ALE) v4<==>v6 signaling translator:(*** click here for elaborate view)

- Involved in this development and maintenance of this signaling translator.Application was built from the scratch with C/C++ and Bash scripting as the backbone to run over the Linux Operating system.
- Application was designed to run over a virtualized environment(likes of VmWare,KVM,HyperV.)

(UB-CSE) Server-client webchat application:(*** click here for elaborate view)

- As an academic project designed a full fledged Server-Client chat application using C++ and prominent networking protocols like TCP,UDP which can act as a replica of WhatsApp web.
- CPP data structures like arrays,lists and pointer manipulations prominently leveraged to achieve the goal here.

(UB-CSE) LUCID Vending - Mobile Application:(*** click here for elaborate view)

- This cross-platform(Android/ios) application was built for a client requirement as part of the Master's Project development course.
- Requirement is to build a full fledged online payment application specifically targeting the vending machine sales including provision for other utility functionalities as well.

- Tech stack included React-Native,Tailwind CSS, Javascript, Firebase, StripeJS for payment related functionalities.

(UB-CSE) Minor Academic Projects -Techstack -Python,OpenCV,Solidty,NodeJS.(*** click here for elaborate view)